

ATRIAL FIBRILLATION WITH RVR —

Clinical One-Pager

Irregularly irregular, narrow QRS, no P waves, rate >100 · 2023 ACC/AHA/ACCP/HRS · IM resident reference

1 · RECOGNIZE

AF w/ RVR = irregularly irregular rhythm · narrow QRS (<120 ms) · no discernible P waves · ventricular rate >100. Atria fire ~400 bpm; **AV-node refractory period** gates the rate (→ AV-nodal blockers work).

Wide-complex AF only with underlying BBB or rate-related aberrancy — irregular irregularity still points to AF.

2 · WORK-UP (new AF)

Universal: TTE, BMP + Mg, TSH (± reflex T4), CBC, ECG ± troponin. Order the rest in context.

Cardiac	Non-cardiac
Valvular / structural HD; decompensated HF (±BNP); ischemia (trop)	Pulmonary/PE (±CXR); OSA screen; drugs/EtOH (±tox); electrolytes; hyperthyroid

3 · STABILITY FIRST

UNSTABLE?	Action
YES	SBP <90, or SBP↓ >30 + [CP / AMS / SOB] from rhythm → synchronized DCCV 120-200 J (sedate if able, don't delay) + anticoag
NO	Rate control to HR <110 . Acute HR >130 → IV meds; HR 110-130 → PO meds

Lenient HR <110 acceptable if asymptomatic + preserved EF (RACE II).

4 · RATE-CONTROL AGENTS

Agent	Line / dose / caution
Metoprolol (BB)	1st; IV 5 mg q5min ×3, PO 12.5-37.5 q6h; ↓↓BP, caution decomp HFrEF/sepsis
Diltiazem (CCB)	1st; IV 10-20 mg bolus+drip, PO 30-90 q6h; ↓↓BP, AVOID HFrEF
Digoxin	2nd; IV 0.25-0.5 mg load → 0.25 q6h; BP-neutral + inotropy; caution elderly/renal
Amiodarone	2nd; IV 150 mg bolus + drip; favored in shock; caution chronic AF (may cardiovert); lung/liver/thyroid tox

Fail one agent → add a second class (e.g., CCB on top of BB).

SCENARIO PICKS

- HFrEF:** avoid CCB; use BB once compensated; digoxin/ amiodarone OK
- Septic shock / hypotension:** amiodarone (most BP-neutral); digoxin alt
- BP-limited after BB:** digoxin (BP-neutral + inotropy)
- Unstable / decompensating:** synchronized DCCV

5 · ANTICOAGULATION

Cardioversion + AF ≥48 h or unknown duration: ≥3 wk therapeutic anticoag before OR TEE to exclude thrombus → then **≥4 wk after, regardless of score** (atrial stunning).

Long-term stroke prophylaxis by **CHA₂DS₂-VASc** (OAC if ≥2 M / ≥3 F). 2024 ESC: **CHA₂DS₂-VA** (drops sex). Anticoagulate everyone cardioverted.